Apex GP — The Ultimate F1 Management Mobile Game Production Document

Version 1.0 · April 2026 · Solo-developer production bible

Apex GP is the F1 management game the mobile market has been waiting for since Motorsport Manager Mobile 3: a deeper simulation than Motorsport Manager 4, a fairer economy than F1 Clash, the narrative drama F1 Manager never delivered, and the multiplayer depth only iGP Manager ever attempted. It ships unlicensed (fictional teams, drivers, circuits) to bypass FOM/FIA constraints and to enable the historical-era, multi-tier, and modding features the licensed games cannot. The stack is React 18 + TypeScript + Vite + Tailwind v4 + Zustand 5 + Motion 12 + shadcn/ui + Capacitor 7, free-to-play with ethical cosmetics-biased IAP. Target retention: **D1 40% · D7 15% · D30 8% · ARPDAU $0.20 at launch**.

The document is organized into eight parts. Part 1 is a mechanical autopsy of the Motorsport Manager series. Part 2 is the competitive landscape and unfilled demand list. Part 3 is the real F1 rules cheat-sheet. Part 4 is live-ops and F2P economy design. Part 5 is the technical architecture blueprint. Part 6 is the complete fictional universe. Part 7 is the full Game Design Document with formulas. Part 8 is the Claude Code prompt library.

PART 1 — Motorsport Manager mechanical deep dive

1.1 Mechanic evolution across MM1 → MM4

MM Mobile 1 (2014, £5.99) was deliberately shallow: three bundled research branches (Aerodynamics/Design/Manufacturing), two-to-three dry compounds without an explicit temperature model, no ERS, no engine modes, single-session qualifying with setup sliders, four tiers of single-seater racing. Monetization was a clean one-time paid model with two optional cash-pack IAPs. TouchArcade called it "relatively unobtrusive and victimless." TouchArcade TouchArcade

MM Mobile 2 (2017, £3.99) introduced the first real depth: **individual car parts** (front wing, rear wing, engine, brakes, gearbox, suspension) with Basic/Good/Excellent component tiers; **engine modes** with fuel-burn multipliers Wikipedia of roughly 0.9× (lean) / 1.0× (standard) / 1.1× (rich) / 1.3× (overtake); five dry compounds Google Play (Ultras/Supers/Softs/Mediums/Hards) with cliff percentages datamined at 20/15/10/5/0%; Motorsport Manager PC W… a tri-branch research tree (Financial/Part/Drivers). TouchArcade Still one-time paid, four IAPs.

MM Mobile 3 (2018, £3.99) is the series peak. Pocket Gamer 4.5/5. Wikipedia Critical additions: **ERS** (Hybrid/Neutral/Power) in selected championships; Level Winner **Mechanics** as a second staff type distinct from Engineers; Google Play Playsport Games a **component-slot/perk system** where a part's bonuses depend on fitting discrete slots within a Skill Point budget (overfilling busts like blackjack); Level Winner **Influence** as a second currency spent on rule-change voting, Playsport Games driver re-signings, and design theft; Team VVV **three categories** of championship

(Open Wheel, GT, Endurance, Team VVV nine sub-championships); Wikipedia Playsport Games **Invitational Races** as annual tentpoles; Google Play Google Play explicit **tyre temperature gauge** with a central sweet spot and wear penalties at both extremes; Steam Solo five driving modes (Attack/Push/Neutral/Conserve/Back-Up) with wear multipliers 1.4/1.2/1.0/0.8/0.6; steamcommunity multi-run qualifying with per-run tyre choice; driver Smoothness stat materially extending stint length Fandom (up to three laps' difference between Smoothness 5 and 20). Steam Community

MM Mobile 4 (Sep 2023, £4.99) is widely derided as **"a carbon copy of MM3"** google (Ruairidh Allan review, 114 helpful votes). New: pre-planned multi-stop **Pit Strategy**; Google Play split **Research vs Development** R&D; **Race Strategist** as a third staff role; Google Play **3D cars on mobile for the first time**; Google Play **Sprint Races + Practice Sessions** added to race weekends; Google Play +2 Hard Mode Update (Nov 2023) added three championships and Stewart Island. App Store But **part-perk attribution UI regressed** (players "unable to tell which part which bonus is coming from"), Google Play Google Play constructor name character cap dropped from 30 → 14, App Store and the leaderboard constructor-name bug App Store from MM3 carried over. App Store Post-launch support effectively ended October 2024. App Store

1.2 The complaint canon

The consolidated MM4 complaint list, mined from App Store, Google Play, TouchArcade, 148Apps, PocketGamer, and r/MotorsportManager:

- **"Carbon copy of MM3"** — the single most-cited complaint.
- **AI regressions**: double-stacking pit stops, pitting in the last three laps, no rain-awareness in qualifying, App Store apple no consideration of tyre delta when calling strategy.
- **Rubber-banding** after two seasons: players who dominate suddenly go from P1 car to P5 "by 1.5 to 2 seconds" with no explanation. apple App Store
- **Race view flickers** and **the track disappears** on some Android devices Google Play (Javin Phung, Aug 2024). google google
- **Paused race with no unpause button**; **tooltip Continue button never appears** App Store (reviewunit, Mar 2026). App Store
- **Constructor name cap reduced** from 30 to 14 characters; App Store old constructor names persist on leaderboard forever. App Store apple
- **Part perks attribution lost**: you can no longer see which part grants which bonus. Google Play
- **No line graph for track wetness**, App Store no interval timing, no rival car stats viewer.
- **Sponsorships** expire with no renewal path; **contracts blindly negotiated** with three rejections locking you out for 90 in-game days. TouchArcade
- **Team member aging curves are wrong** — engineers decline in their 40s, which players flag as unrealistic. App Store
- **No cap on simultaneously-built parts** — "feels cheap." apple

- **IAP proliferation**: MM4 has 10 IAPs vs MM3's 5, including a paid Premium Livery Bundle and paywalled Game Editor at $5.99.
- **Post-launch abandonment** — "a game that hasn't seen updates in a year" (App Store) (Mar 2026). (App Store)

1.3 Features fans have requested forever but never received

Livery editor. Multiplayer (even async). Historical seasons officially supported (mods-only today). Custom championships with player-authored calendars and rules. A real tyre temperature + pressure + camber + track-rubber-evolution model. A proper F1-style cost cap with audits. Power unit allocation (ICE/MGU-K/MGU-H/TC/ES/CE) with grid penalties. Stewards inquiries and team appeals. Press conferences and media sentiment. Team orders beyond swap-positions (fuel-save target, defend-only, pit priority). Voiced pit radio. More motorsport disciplines (NASCAR/MotoGP/rally). (Google Play) An F2/F3 academy pipeline where you can own the junior team and promote drivers into your F1 team. Configurable race lengths beyond Short/Medium/Long. (google) Cloud save. Per-track setup save/copy. In-race gap graphs, lap-by-lap telemetry, and wetness-over-time charts.

1.4 What MM1/MM2 had that MM3/MM4 lost

Qualifying setup sliders (MM2 had manual gear-ratio and aero sliders; (Wikipedia) MM3 replaced them with a mechanic-driven card minigame, (TouchArcade) widely considered a depth regression). Tyre pressure and camber in advanced setup. (Steam Community) Part-perk transparency (MM3 had it clearly; MM4 obscured it). Clean tier-ladder progression (MM1's four-tier WMC ladder was more satisfying than MM3/4's branching multi-discipline structure for single-seater purists). Unchanged soundtrack (minor fatigue). And the most important regression: **post-launch support durations shortened with every sequel**.

1.5 Unpatched MM4 bugs as of Mar 2026

Leaderboard constructor-name persistence. 14-char cap on constructor names. Race starts paused with no buttons. Missing tooltip Continue button. Race screen flickers black on Android. (google) Track disappears mid-race. (google) Support survey text box broken. (google) Rubber-banding after season two. AI double-stacking in last three laps. Achievements intermittently failing to trigger (temp workaround was disabling Achievements and Leaderboards entirely (App Store) in patch 2023.1.9). (apple) Background/foreground lifecycle issues on some Android builds.

PART 2 — The F1 game landscape

2.1 F1 Manager 2022/2023/2024 (Frontier) — dead franchise, massive lessons

Frontier held a four-game exclusive F1 license from 2020. (GamePressure) Only three games shipped. **F1 Manager 2025 was voluntarily cancelled in September 2024**, confirmed in Frontier's FY24 Annual Report (10 Sept 2024) via a £15.0m IP-specific impairment. (RacingGames)

H1 FY24 operating loss was £30.8m versus £6.9m profit H1 FY23. Morningstar Frontier's public reasoning: refocus on its proven Creative Management Simulation genre (Planet Coaster 2, Jurassic World Evolution 3) and exit third-party IP. The four-game contract was expiring anyway; Frontier elected to take the write-off early.

F1M22 (Metacritic ~75): praised for six-part car development, TV-style presentation, Sky Sports commentary, real driver radio; criticized for "hive-minded" AI, broken safety car behavior, no damage model, and Frontier's early-abandonment announcement that was later reversed but never forgiven. RacingGames

F1M23 (evolution over revolution): added Race Replay scenarios, Sporting Director role, Formula 1 driver confidence and tactics dials, nuanced tyre temperature, driver aging, Las Vegas and Losail circuits. Sales dropped 34% vs F1M22 in H1 FY24 reporting. Team-switching promised at launch, delivered only in September 2023 update.

F1M24 (best-reviewed, Metacritic ~80): shipped **Create-A-Team** Metacritic (the long-begged-for feature) with four origin archetypes, engine manufacturer choice, livery/logo customization; Impulse Gamer F1 Manager Mentality Hub; Affiliate Drivers as a partial academy answer; dynamic Marketability attribute; mechanical failures finally added; Switch port. But Steam criticism captured the persistent gap: "When Lance Stroll on 40% hard tires can overtake Max Verstappen on 90% softs, you know something's seriously broken." Metacritic Steam Community

Why Frontier failed commercially is the most important analysis for this project. Five factors compound: (1) annual cadence with minimal year-over-year change, (2) abandoning F1M22 support broke trust, (3) Drive-to-Survive's casual audience wanted narrative drama, not spreadsheets, (4) free Motorsport Manager mods delivered more realism for zero marginal cost, (5) the hardcore audience wanted even *more* depth than Frontier delivered while the casual audience wanted less. **The niche is real but split; you must decide which audience you serve on every design decision.** Apex GP serves the hardcore audience on depth and the casual audience on session length — this is the central design tension the document resolves.

2.2 F1 Clash (Hutch, 2019)

F1 Clash is the category's commercial leader on mobile (~$900k/month on ~100k installs per Sensor Tower estimates). Sensor Tower Its core loop is **6–9 lap head-to-head async races** F1 Manager Wiki lasting about five minutes, with card-based drivers (5 rarities) and six component slots, climbing through 12 Series (Melbourne → Suzuka/Monaco → Baku). Monetization stacks: coin and bucks dual-currency, Hutch crates with timed unlocks or buck-skip, Pit Pass battle pass (free + premium tracks), Series Premium, fusion for dup dumps, Golden Spin daily wheel, direct coin/buck packs.

The **UK ASA upheld three complaints against F1 Clash on 19 November 2025** (Leon Xiao case). The App Store listing's "Offers In-App Purchases" notice was ruled insufficient disclosure of loot boxes (CAP Code 3.1/3.3). CMS Lawnow The Golden Spin wheel's 12 equal-spaced icons were ruled misleading because actual probabilities varied significantly and the "?" info icon was deemed inadequate signposting. The "TRY NOW FOR UP TO X10 BETTER PRIZES" claim was *not* upheld (data substantiated it). OpenCritic The enforcement outcome is that any unlicensed

F1 mobile game targeting UK players must publish drop rates clearly, cannot visually imply equal probability on differential-odds reward surfaces, and must disclose loot-box mechanics immediately adjacent to IAP callouts in store listings. ASA

Player sentiment: loved for pickup-and-play, head-to-head, and live Grand Prix events tied to real race weekends; Google Play hated for **Team Score matchmaking that creates a win/loss trap**, a Series 5–6 pay-to-win wall, Pit Pass timing complaints, RNG inconsistency, and the community-accepted reality that Hutch offers in-game gifts for 5-star reviews (inflating store ratings).

2.3 iGP Manager (2011, still live)

The longest-running F1 management MMO. Cross-platform browser + iOS + Android + Mac + visionOS. App Store Real-time multiplayer races up to 32 managers per league. App Store Real-world AR weather streaming Browser Craft (Monaco weather in-race = actual Monaco sky). Endorsed by Lando Norris and Fernando Alonso. App Store **iGP's core insight is that live human opponents generate the drama no single-player sim can script.** Its weaknesses are a dated UI, sparse data exposure, only-next-stint strategy adjustment, Igpmanager and race-view pop-overs that prevent following track and strategy simultaneously. Igpmanager Monetization is ethical (subs are "comfort and fun rather than real advantage" Igpmanager per community); IAP primarily cosmetics.

2.4 Motorsport Manager PC (2016) and its mod ecosystem

The PC version delivers the depth the mobile series cut: individual aerodynamic components with risk/reliability/weight tradeoffs, suspension and brake-tuning screens, rulebook voting on spec engines/refueling/driver-weights/calendars, Playsport Games a Spy Network for rival intel, tyre supplier selection, random-event dilemmas, and four tiers of championship culminating in the World Motorsport Championship. **What is most revealing is the mod community**: Ultimate F1 Retro Season Mods Steam Community (1995–2002 Steam Community real drivers and liveries with reworked super-licence), F1 Manager 2021– Steam Community2026 mods adding F1+F2+F3+IndyCar+WEC and 65+ circuits, the Fire Fantasy Mod reworking tyre physics. **The mods prove what players want: historical seasons, multi-series pipelines, rebuilt reliability and parts, pre-season testing, super-licence gates, rule-change voting, real liveries/sponsors, aggressive AI.**

In February 2026 Playsport reclaimed MM rights from Sega and released Patch 1.6 with Apple Silicon support — a sequel is hinted but unannounced.

2.5 Unfilled community demand — the design opportunity

Consolidated from Reddit, Steam, TouchArcade, and Frontier forums: **deep narrative career story** (Drive-to-Survive analog — Codemasters' Braking Point is the closest reference but it is a driving game not a management game); **realistic driver market** with gardening leave, mid-season swaps, personality-driven negotiations; **multi-generational careers** with rookies aging into veterans and retiring into team-principal roles; **power unit manufacturer management** (develop ICE/MGU-K/MGU-H separately, sell engines to other teams, engine-freeze handling); **technical regulation changes** between seasons (ground-effect transitions, active-suspension

bans, spec engine periods); **full livery editor** plus sponsor placement control; **proper cost cap** with FIA-style audits and penalties; **F2/F3 academy integration** where you can own junior teams and promote to F1 ("Zero games deliver this" — Jump Dash Roll); **async multiplayer leagues**; **modding support** (the absence of which kills Frontier's longevity); **reactive live race commentary** beyond "Copy"; **weather radar** with rain-in-N-laps visualization and **proper damage models**.

PART 3 — Real F1 rules cheat-sheet

3.1 Tires (Pirelli, 2024–2026)

Pirelli produces six slick compounds C0 (hardest) through C5 (softest). Wikipedia Formula 1 For each Grand Prix weekend Pirelli selects three: the Hard (hardest of the three), Medium, and Soft. Formula 1 PlanetF1 Intermediates are a single compound for damp/drying conditions, Full Wets for standing water. Formula 1 Weekend allocation is 13 sets for a normal weekend Formula 1 (8 soft, 3 medium, 2 hard typical), Autosport reduced for Alternative Tyre Allocation sprint weekends. A dry race must use at least two different compounds. Substack +2 Operating windows sit roughly at 100°C optimum (surface) for Soft, 95°C Medium, 90°C Hard, with a roughly 15°C sigma gaussian grip response. Intermediate optimum ~55°C, Wet ~40°C. Real lap time deltas roughly Hard→Medium = 0.5–0.8s, Medium→Soft = 0.4–0.7s, PlanetF1 varying by circuit. Degradation is linear with a cliff: Soft cliff ~15 laps, Medium ~22, Hard ~32; cliff drop ~0.08s/lap² beyond threshold. **2026 regulation**: narrower tyres and revised construction, still Pirelli-supplied. Athlon Sports Formula 1

3.2 Pit strategy and race operations

Pit lane speed limit is 80 km/h at most circuits Motorsports-Regulations.c... (60 km/h at Monaco/Melbourne/Singapore in specific sessions). Motorsport Total pit loss green-flag varies 18–25s depending on pit lane length; F1 Chronicle Themotorsportmetrics Monaco 18.5s, Race Sundays Singapore 27s. Physical stop time ranges 2.3–3.1s team-dependent (Ferrari fastest 2.44s mean in 2025, Haas/Williams 3.05s). F1pace **Undercut**: pit one lap earlier than rival, Steam Community exploit new-tyre pace GPfans to gain Motorsport 0.5–0.8s/lap for 2–4 laps, typically gaining 1.5–3s of track position, works best on high-deg compounds/tracks. GP Explained +2 **Overcut**: stay out, Steam Community rival on fresh tyres is blocked by traffic or loses tyre temperature out-lap, GPfans you gain relative pace, Motorsport works on low-deg tracks. GP Explained +3 **Safety car pit math**: field is running at ~40% pace behind the SC, so the pit lane delta collapses to ~5–8s total loss ("free stop"). Motorsport **Virtual Safety Car**: ~55% of SC benefit, ~12s pit loss. **Double-stacking** incurs +3–5s penalty on the second car (wait time + reset).

3.3 Cost cap and development

Current cost cap is **$135M** (2024, indexed), excluding driver salaries, top-three-staff salaries, marketing, and engine development (power unit cap is separate). GPblog CapEx cap is ~$36M over four years. **Aerodynamic Testing Restrictions (ATR)**: wind tunnel hours and CFD hours scale inversely with Constructor Championship standing — the 1st-place team gets 70% of a

baseline allowance, 10th-place 115%. Motorsport Medium Mandatory 14-day summer factory shutdown Atlassian Williams Racing plus winter shutdown. Medium Parts are homologated at season start for safety-critical components; aero parts can be upgraded throughout the season under the cap.

3.4 Power unit

Six components: ICE (Internal Combustion Engine), F1 Chronicle Turbocharger (TC), MGU-H (Motor Generator Unit - Heat), MGU-K (Motor Generator Unit - Kinetic), Energy Store (ES, the battery), Control Electronics (CE). Autoevolution Formula 1 Season allocations per driver for a 24-race calendar are typically 4 ICE, 4 TC, 4 MGU-H, 4 MGU-K, 2 ES, 2 CE. F1-Fansite Exceeding allocation triggers grid penalties: Flow Racers 10 places for first additional ICE/TC/MGU-H, 5 for the second; F1 Mavericks Dive Bomb 10 for first MGU-K/ES/CE; pit lane start for exceeding beyond that within same race. Gearbox cycles: 8 per season, 5-place penalty for early change.

ERS constants: ES capacity 4 MJ, max deploy ES→MGU-K per lap 4 MJ, max harvest MGU-K→ES per lap 2 MJ Autosport Distantreader (MGU-H flows uncapped, Quora enabling effective harvest beyond 2 MJ via H-to-K direct). Somersf1 At full deployment, ERS is worth roughly 0.5s/lap (120 kW over ~33s of straights). Flow Racers Distantreader

2026 regulation: MGU-H eliminated, McLaren Racing MGU-K boosted Wikipedia to 350 kW Medium (versus today's 120 kW), GitHub +4 ICE power reduced so electric/ICE split approaches 50/50, McLaren Racing Top Racing Shop fully synthetic fuel, Medium active aero replaces DRS with an "Overtake Mode" electrical override. Sportsnaut Sky Sports

3.5 DRS, driver market, qualifying, stewarding, points

DRS: activation requires being within 1.0s of the car ahead at the detection point; GPfans Wikipedia enabled after lap 2; GPfans disabled in wet conditions. F1 Chronicle +2 Lap time benefit 0.1–0.3s per zone, Formula1 0.5s/lap aggregate on multi-zone circuits Wikipedia (pre-2026).

Driver market: contracts 1–3 years typical; silly season announcements cluster May–September. Academies: Ferrari Driver Academy, Red Bull Junior Team, Mercedes, Alpine, McLaren. **Super Licence** requires 40 points over three years, with F2 champion getting 40 points and runner-up also 40. Pay drivers historically brought $10–30M/year sponsorship (Stroll, Mazepin, Latifi references).

Qualifying format: Q1 18 minutes with bottom 5 eliminated → Q2 15 minutes with bottom 5 eliminated → Q3 12 minutes for top 10 pole shootout. RacingNews365 Wikipedia Parc fermé engages at the start of Q1; no setup changes until race. The Race The Q2-tyre-starts-race rule was repealed.

Stewarding: Safety Car triggers on debris, stranded cars, significant weather. Red flag on more serious hazards; standing restart after ≥2 lap stoppage. Track limits: white-line rule with three-warning-strikes before lap time deletion plus 5-second penalty. Motor Sport Magazine Penalty ladder: 5-second → 10-second → drive-through (~20s) → stop-go (~30s) → grid penalty → disqualification. **Penalty points** accumulate on a Super Licence; 12 points within 12 months triggers a 1-race ban.

Points: 25-18-15-12-10-8-6-4-2-1 for P1–P10. Fastest lap point (must finish top 10); may be removed for 2025+. Sprint race: 8-7-6-5-4-3-2-1 for P1–P8. Constructor championship sums both drivers.

PART 4 — Mobile F2P design and live-ops

4.1 Retention and monetization targets

Mid-core strategy benchmark retention is D1 35% / D7 10% / D30 6% industry average; top-quartile is D1 40%+ / D7 15%+ / D30 8–12%. Apex GP targets top-quartile at global launch. Session count 2.5–4/day, session length 12–25 minutes, DAU/MAU 20%+. Mid-core payer conversion 2–5% industry, top-quartile 5–8%. ARPDAU $0.15–$0.40 median mid-core, top $1.00+; Apex GP targets $0.20 at launch ramping to $0.35 at live-ops maturity. Whale share: top 10% of spenders deliver ~70% of IAP revenue; whale ARPPU ~$335/month. Apex GP caps whale yield per dollar by diminishing-returns VIP tiers plateauing at $200 lifetime spend.

4.2 Economy structure

Dual currency: **Prize Money** (soft, grindable) and **Tokens** (hard, purchasable but also earnable ~150/day free). IAP ladder: $0.99 impulse, $4.99 starter, $9.99 battle pass, $19.99 premium pass, $49.99 whale currency bundle, $99.99 top tier. **Battle Pass** ("Racing Pass") at $9.99 for 45-day seasons, free + premium tracks, premium earnback in Tokens exceeds purchase price (Fortnite model). A $19.99 "Championship Pass" adds cosmetic-only tier skips; no power skips. **Starter pack** at $4.99 is triggered only after session-one victory, valid 72h, delivers a livery skin + Tokens + R&D boost.

Crate/gacha rules (ASA-compliant, published rates): soft pity starts at 70% of the way to hard pity; hard pity guarantees featured driver at 80 pulls; 50/50 featured mechanic with fate-points carryover; all drop rates published in-store and adjacent to every purchase surface; loot box disclosure in store listings immediately next to IAP callouts.

4.3 Live-ops calendar and real-race hooks

Race Weekend Live Events hook Apex GP's in-game calendar to real F1 race weekends at abstract level (we do not use real track names). Friday: practice challenges (hit target lap times, ad-unlock for extra attempts). Saturday: qualifying tournament bracket. Sunday: **Ghost Race** where your AI car races a deterministic ghost seeded by the event, leaderboard closing when the real race ends. GameRefinery data shows time-limited events drive 20–40% DAU spikes on event day.

Weekly cadence: Monday driver market refresh, Wednesday mid-week tournament, Friday weekend event kickoff. Monthly cadence: Battle Pass rotation. Quarterly tentpoles: season opener, mid-season Principality Circuit (Monaco analog) showcase, summer break update, season finale championship week.

4.4 Push notifications and soft launch

Push opt-in prompt is deferred until after tutorial race win (Pushwoosh case study achieved 97% opt-in with pre-permission screen and clear value). Frequency soft cap 1/day, hard cap 5/week. Contextual notifications: "Race starts in 1 hour" (real calendar), "Pit crew finished upgrades," "Rival team overtook you in league," "AFK chest is full," "Battle Pass ends in 48 hours." Urban Airship data: push drives +120% 90-day retention.

Soft launch path: Phase 1 Philippines + Indonesia, Android-only, 2–3 months, focus on D1/tutorial completion/crash rate/CPI. Phase 2 adds Netherlands, Ireland, New Zealand on Android + iOS, 2–3 months, focus on D7/D30/ARPDAU/IAP conversion/Pass attach. Phase 3 adds Canada, Australia, Nordics, 1–2 months, focus on LTV and ROAS payback. Global gate: D1 ⩾ 35%, D7 ⩾ 10%, D30 ⩾ 5%, payer conversion ⩾ 2%, ARPDAU ⩾ $0.15.

PART 5 — Technical architecture

5.1 Stack snapshot (April 2026)

React 18.3 (not 19 — Motion 12 ecosystem stability). TypeScript 5.5+. Vite 6.x. Tailwind CSS 4.0 GA with Lightning CSS. Zustand 5.x (useShallow from `zustand/react/shallow`; `createWithEqualityFn` from `zustand/traditional`). Motion 12.x (`motion/react` import). shadcn/ui generated with Radix primitives + Tailwind v4. Capacitor 7.6.x (iOS 14+ deployment target, Android minSdk 23 targetSdk 35, Xcode 16, Node 20, JDK 21). @revenuecat/purchases-capacitor 11.3.x. @tanstack/react-virtual 3.x. Zod 4.x.

5.2 Zustand at scale (10k+ entities)

Single bound store with typed slices via `ImmerStateCreator<T>` helper. Immer enabled with `enableMapSet()`. Middleware order `devtools(subscribeWithSelector(persist(immer(…))))`. Entities normalized as `Map<Id, Entity>` with a parallel `Id[]` for iteration order. List components subscribe to `driverIds` only; each row subscribes to `useGameStore(s => s.drivers.get(id))` and wraps in `React.memo`, so a bulk tick updating 20 drivers re-renders ~20 row components out of ~200 visible rows, not all 10,000.

Multi-value picks require `useShallow(s => ({money: s.money, rep: s.reputation}))` or Zustand 5 infinite-loops. Transient per-frame values (speedometers, HUD ticks) use `useMotionValue` outside React and bypass the store entirely. For hot sim loops that update 10k entities, bypass Immer and mutate the Map in place — returning `{drivers: state.drivers}` from `set()` drops tick cost from ~15ms to ~2ms on a mid-range phone.

Derived state tiering: cheap (<1μs/entity) in selectors; medium in component `useMemo`; expensive cross-cutting computed once in the reducer and stored on `state.derived.*`.

5.3 Rendering optimization

Virtualize with `@tanstack/react-virtual` for any list >50 items (drivers, parts, historical races); overscan 8, estimateSize dynamic via `measureElement`. Use `useTransition` for route changes and leaderboard filters; **never wrap the simulation tick in a transition** — transitions can be dropped, which breaks determinism. Use `useDeferredValue` for text-driven filters. React Compiler 1.0 (October 2025) is evaluated opt-in but not required; if adopted, strip manual `useMemo`/`useCallback`.

Motion 12 rules: animate only `transform` and `opacity`; never enable `layout` on race-tick list rows; use `LazyMotion` with `domAnimation` features to drop ~30KB; throttle animations during season fast-forward via `<MotionConfig reducedMotion="always">`.

5.4 Deterministic simulation

PRNG is **sfc32** seeded via xmur3 hash of a string seed. Split streams per subsystem: `race`, `ai`, `damage`, `events`, each seeded `masterSeed + ':' + name`. sfc32 state is `(a,b,c,d)` serialized into save. ESLint `no-restricted-properties` bans `Math.random`. Test: run season twice with same seed, deep-equal final state.

5.5 Save versioning, storage, analytics, billing, submission

Zod schemas version-gated with a `_version` literal. Migrations array from version N to N+1 applied sequentially at load; final schema validated with `SaveV7.parse(data)`. Storage: `@capacitor/filesystem` for save blobs (hybrid copy-bak, write-tmp, rename pattern); `@capacitor/preferences` for settings/metadata. Avoid localStorage/IndexedDB — WKWebView purges under pressure. Cloud save via custom backend keyed to RevenueCat app user id.

RevenueCat integration uses entitlement-based gating (check `customerInfo.entitlements.active['pro']`, never product ids); **RevenueCat Test Store** (2025) for sandbox-less testing; Restore Purchases button required (Guideline 3.1.1). Analytics via GameAnalytics community plugin or Firebase Analytics + BigQuery; event taxonomy covers `session_start`, `race_completed`, `upgrade_purchased`, `iap_started/completed`, `funnel_onboarding`, `save_corrupted`. **Do not link ATT framework unless needed** — Guideline 2.1 auto-rejection. If ad SDKs present, pre-prompt before ATT for 50–70% opt-in vs 25–35% cold.

iOS submission 2025: Xcode 16, iOS 14 minimum, mandatory PrivacyInfo.xcprivacy with required-reason API codes (UserDefaults CA92.1, file timestamp C617.1/0A2A.1). **Google Play 2025**: target SDK 35 by Aug 31, 2025; Play Billing Library 7.0+; Data Safety form must match SDK behavior; 14-day closed testing with 12 testers required for new personal developer accounts (reduced from 20 Dec 2024).

PART 6 — The fictional universe

Apex GP ships unlicensed. All teams, drivers, circuits, sponsors, and suppliers are original creations. This section is the canon reference for narrative, livery, and content generation.

6.1 The ten teams

1. **Albion Grand Prix (UK, Woking)** — Storied British midfielder-turned-contender. Racing green/orange livery. Strength: aero (wind tunnel legacy). Weakness: engine (customer spec). Principal: Sir Miles Hargrove (patrician, PR-savvy). Backstory: founded 1964, 8 constructors' titles across decades, currently emerging from a decade in the wilderness.
2. **Scuderia Veloce (Italy, Maranello analog — Emilia)** — Heritage aristocracy. Scarlet livery, yellow shield. Strength: brand, chassis, engine (works). Weakness: political infighting. Principal: Donna Carla Rosselli. Backstory: only team to have competed every season of the championship; 15 constructors' titles; fans riot when they don't win.
3. **Königsbergring Motorsport (Germany, Stuttgart)** — Ruthlessly methodical. Silver/turquoise. Strength: reliability, operations, engine (works). Weakness: culture (rigid, slow to pivot). Principal: Dr. Klaus Weiss.
4. **Stellare Racing (Austria, Spielberg)** — Energy-drink analogue with a driver academy. Navy/red. Strength: aero, young-driver pipeline. Weakness: internal politics. Principal: "The Advisor" Dieter Krauss.
5. **Mistral (France, Enstone analog — Normandy)** — Works French manufacturer effort, nostalgic yellow/blue. Strength: chassis. Weakness: strategy. Principal: Florent Dubois.
6. **Sakura Kasei (Japan, Sakura)** — Precision engineering, works engine. White/red rising sun motif. Strength: engine, reliability. Weakness: aero. Principal: Hiroshi Tanaka.
7. **Caldera Racing (Switzerland, Hinwil)** — Independent outfit, customer Veloce engine. Blue/white. Strength: rookie development. Weakness: budget. Principal: Beatrice Müller.
8. **Meridian Motorsport (USA, Charlotte + Silverstone dual HQ)** — American brash ambition. Stars/red/navy. Strength: chassis. Weakness: operations. Principal: Chuck Halloran.
9. **Aurora GP (India, Mumbai tech-corridor)** — Rising outsider, purple/gold. Strength: software, driver aids. Weakness: manufacturing. Principal: Priya Desai.
10. **Borealis Racing (Finland, Helsinki)** — Rally-heritage, works Nordic engine. White/blue-arctic. Strength: wet weather, cold tyre management. Weakness: summer hot tracks. Principal: Mika Järvinen.

6.2 The twenty drivers (two per team)

Each has **8 stats on 0–100 scale**: Pace, Cornering, Braking, Smoothness, Consistency, Overtaking, Defending, Wet Skill; plus Marketability, Feedback, Adaptability, Morale, Age, Nationality, and 1–3 Traits.

Elite tier (85+ overall): **Luca Moretti** (Veloce, 28, Italian, flamboyant, tire-shredder but magic in wet); **Max Brandt** (Königsbergring, 31, Austrian, robotic precision, poor starts); **Jack Calloway** (Albion, 24, British, ice-cold strategist, rising star); **Samuel Okonkwo** (Stellare, 26, Nigerian, academy graduate, overtaking savant); **Tomás Silva** (Mistral, 32, Portuguese, veteran smooth-operator).

Upper midfield (78–84): **Kenji Nakamura** (Sakura Kasei, 29, Japanese, outstanding feedback); **Viktor Kovač** (Caldera, 23, Croatian, rookie phenom with hot temper); **Sebastien Laurent** (Mistral, 27, French, Mistral academy lifer); **Riley Jameson** (Meridian, 25, American, crash-prone talent); **Arjun Varma** (Aurora, 28, Indian, wet-weather specialist).

Midfield (70–77): **Oliver Hartwell** (Albion, 34, British, ageing veteran, great mentor trait); **Dmitri Volkov** (Königsbergring, 22, Russian, pay driver with real talent); **Hugo Reinholt** (Stellare, 30, German, reliable #2); **Chika Agbaje** (Meridian, 24, Ghanaian-American, marketability star); **Elena Costa** (Caldera, 26, Spanish, first female championship contender in canon).

Back of grid (60–69): **Ivar Lindqvist** (Borealis, 33, Swedish, wet-skill specialist); **Matti Virtanen** (Borealis, 21, Finnish rookie); **Carlos Fernández** (Aurora, 35, Mexican, pay driver veteran); **Hans Richter** (Sakura Kasei, 34, German, second driver, retiring); **Naoko Suzuki** (Aurora, 22, Japanese, first female Asian champion aspirant).

6.3 The ten circuits

Each has: length (km), laps, type (street/permanent), DRS zones, SC probability, rain prior, weight vector for car performance.

1. **Principality Circuit** (Monaco analog) — 3.34km, 78 laps, street, 1 DRS zone, SC 60%, rain 25%, weights chassis 0.40 / aero 0.30 / brakes 0.20 / engine 0.05 / reliab 0.05.
2. **Autodromo di Bellagio** (Monza analog) — 5.79km, 53 laps, permanent, 2 DRS, SC 20%, rain 15%, weights engine 0.45 / aero 0.20 / brakes 0.20 / chassis 0.10 / reliab 0.05.
3. **Silverthorne Circuit** (Silverstone analog) — 5.89km, 52 laps, permanent, 2 DRS, SC 20%, rain 35%, weights aero 0.40 / chassis 0.20 / engine 0.20 / brakes 0.15 / reliab 0.05.
4. **Spa-Ardennes International** (Spa analog) — 7.00km, 44 laps, permanent, 2 DRS, SC 25%, rain 40%, weights engine 0.35 / aero 0.30 / chassis 0.15 / brakes 0.15 / reliab 0.05.
5. **Suzusaki Raceway** (Suzuka analog) — 5.81km, 53 laps, permanent, 1 DRS, SC 25%, rain 33%, weights aero 0.35 / chassis 0.30 / engine 0.20 / brakes 0.10 / reliab 0.05.
6. **Jeddah Bay Street** (Jeddah analog) — 6.17km, 50 laps, street, 3 DRS, SC 45%, rain 1%, weights engine 0.30 / chassis 0.25 / aero 0.20 / brakes 0.20 / reliab 0.05.
7. **Desert Dunes Autodrome** (Bahrain analog) — 5.41km, 57 laps, permanent, 3 DRS, SC 15%, rain 2%, weights engine 0.30 / brakes 0.25 / aero 0.20 / chassis 0.20 / reliab 0.05.
8. **Marina Heights** (Singapore analog) — 4.94km, 62 laps, street night race, 3 DRS, SC 70%, rain 35%, weights chassis 0.30 / aero 0.25 / brakes 0.25 / engine 0.15 / reliab 0.05.

9. **Interbrasília** (Interlagos analog) — 4.30km, 71 laps, permanent, 2 DRS, SC 40%, rain 45%, weights engine 0.30 / aero 0.25 / chassis 0.20 / brakes 0.20 / reliab 0.05.
10. **Yas Lagoon** (Abu Dhabi analog) — 5.28km, 58 laps, permanent, 2 DRS, SC 10%, rain 1%, weights aero 0.30 / engine 0.25 / chassis 0.20 / brakes 0.20 / reliab 0.05.

6.4 Calendar, sponsors, suppliers, narrative engine

Calendar runs 10 races March–November with two triple-headers, one summer break (mandatory factory shutdown), two sprint weekends (rounds 4 and 8). **Sponsor universe**: 40+ fictional brands across three tiers (Title ⩾ $20M, Major $5–20M, Minor < $5M) spanning automotive (Lumen Oils, Velox Tyres), tech (ZenithCloud, Neural Dynamics), consumer (Orbit Energy, Strata Watches), finance (MonarchBank, Redshift Capital), travel (Stratosphere Airlines), and lifestyle brands. Each has engagement requirements, performance bonuses, and brand values that affect renegotiation.

Parts suppliers: four engine manufacturers (Veloce Works, Königsbergring Works, Sakura Kasei Works, Mistral Works + customer deals) with distinct power/reliability/fuel-economy curves; two tyre suppliers (Velox and Draft — Velox has wider operating windows, Draft is aggressive compound with sharper cliffs) with a mid-season rulebook vote possible; fuel suppliers (Lumen, Prometheus, Helios) with octane/detonation tradeoffs.

Narrative engine generates drama via a rules-based event system seeded per season. Rivalries form when two drivers within same team or adjacent teams collide or overtake each other ⩾3 times in 5 races; crashes escalate inter-driver tension meters. Controversies trigger from: pit lane incidents, team orders refused, cost cap breaches (probabilistic audit), off-track photos of drivers. Team orders produce morale events — a driver who refuses "Let him pass" gets +10 independence but -15 relationship with team principal; consistent obedience builds team chemistry bonuses to pit crew speed and strategy accuracy. Every race weekend generates 3–7 narrative beats surfaced as News/Inbox items.

PART 7 — Game Design Document

7.1 Race simulation algorithm

The core lap time formula is additive:

```
lapTime = baseLapTime
        + tyreDelta(compound, age, track)
        + fuelDelta(fuelKg, track)            // 0.033 s/kg
        + ersDelta(mode, soc, track)           // -0.5 to +0.2
        + driverDelta(driver, track)           // -1.0 to +1.25
        + weatherDelta(weather, compound)      // 0 to +25
        + trafficDelta(gap, track)             // 0 to +0.7
        + drsDelta(eligible, zones, track)     // -0.1 to -0.6
        + noise(driver.consistency)            // N(0, σ)
```

Noise σ = max(0.05, 0.30 × (1 − consistency/100)). Mistake Bernoulli per lap: p = 0.002 × (1 − accuracy/100), cost 1–4s. Driver delta = (75 − overall) × 0.05 where overall = 0.4 pace + 0.25 cornering + 0.15 braking + 0.10 smoothness + 0.10 adaptability.

Car aggregate score weighted per circuit as in Part 6.3; scoreToLapOffset = (50 − carScore) × 0.04, producing a 2–3s best-vs-worst spread.

7.2 Tyre model

Five slicks (C1 Hard → C5 Hyper) plus Intermediate and Wet. Each has `{baseDeg, cliffPoint, cliffFactor, maxLife, warmupLaps, optTemp}`:

- C5 Hyper: 0.09/8/0.10/15/1/102°C
- C4 Soft: 0.08/15/0.08/22/2/100°C
- C3 Medium: 0.05/22/0.06/32/2/95°C
- C2 Hard: 0.03/32/0.05/45/3/90°C
- C1 SuperHard: 0.025/40/0.04/55/4/85°C
- Intermediate: 0.10/18/0.10/28/1/55°C
- Wet: 0.08/25/0.08/35/1/40°C

`tyreDelta = warmupPenalty + baseDeg × age + cliffFactor × max(0, age − cliffPoint)²`, multiplied by driver Smoothness modifier (1.15 − 0.15 × smoothness/100). Tyre surface temperature modeled as two-state ODE (carcass + tread) heated by slip and cooled by airflow; grip = $\exp(-(T-optTemp)^2 / (2\sigma^2))$ with σ = 15°C. Graining accumulates when cold-tire slip exceeds threshold; blistering when hot; both add persistent penalty until cleaned by cautious laps.

7.3 Weather system

Four-state Markov chain {Sunny, Cloudy, LightRain, HeavyRain} stepped per minute with diagonal stickiness ⩾ 0.95. Off-diagonal onset probabilities scaled by circuit rain prior / 0.25. Track wetness is a scalar 0–100%: `dWetness/dt = 8 × rainIntensity − (0.05 × (Tair + Ttrack) + 0.2 × wind + 0.15 × carsOnTrack) %/min`. Crossovers: slicks usable < 15%

wetness, intermediates 10–70%, wets > 50%. Lap time penalty scales with wetness×compound mismatch. Forecast shown with growing uncertainty: σ = 0.05 + 0.01 × minutesAhead.

7.4 ERS, fuel, DRS, SC/VSC, pit stops

Already specified in Part 3 and the simulation research. Five ERS modes (`none`, `harvest`, `balanced`, `overtake`, `hotlap`) with lap time deltas from +0.20 to −0.50 and SoC step updates. Fuel modes (lean/standard/rich/overtake) with burn multipliers 0.9/1.0/1.1/1.3 and lap deltas +0.4/0/−0.2/−0.45. DRS 0.15s per zone with 0.85 stacked-diminish factor. SC probabilities per-circuit with per-lap hazard h = 1 − (1 − P_race)^(1/N_laps). Pit stationary time gaussian(team.mu, 0.30) with 0.5% wheel-gun failure adding 3–10s. Pit lane delta per circuit tabulated; SC factor 0.40, VSC factor 0.55.

7.5 Car development and R&D

Each car has six upgradeable parts (FrontWing, RearWing, Chassis, Underfloor, Sidepods, Suspension) with five stat axes (Downforce, DragEfficiency, MechanicalGrip, BrakingStability, Reliability). **Every part has**: design capacity (Skill Points budget), reliability penalty if over-budgeted, build time (1–6 in-game weeks), cost (soft currency + % of cost cap), carryover probability across seasons. **Component slot system** from MM3 restored: fitting three Good components into a part budget unlocks a perk (e.g., "Tyre Thermal Manager: -10% surface temperature swing").

R&D tech tree has three branches (Aero, Powertrain, Operations) with 30+ nodes each, unlocked via Research Points accumulated via facilities and wind-tunnel/CFD hours. Nodes have risk/reward: high-risk nodes give large deltas but 15–25% chance of "dead end" wasted research. **Cost cap**: $135M/year, driver salaries and top-3-staff excluded; mid-season FIA audit probability 10% with discovery probability proportional to overspend. ATR: wind tunnel and CFD hours scale with Constructor standing.

7.6 Driver stats, development, morale, rivalries

Eight core stats (Part 6.2) evolve via training (HQ-dependent rate), race experience (stat-XP per race finish), and age curves (peak 27–30, decline 33+ with regen probability based on Feedback stat). Morale is a −100..+100 scalar updated by: race result vs expectation, team orders compliance, public comments, off-track events, teammate interactions. Morale ⩾ +50 gives +0.1s lap time; ⩽ −50 gives +0.2s plus mistake rate +25%. **Rivalries** instantiate when two drivers collide or overtake each other ⩾3 times in 5 races; rivalry state increases overtake probability but also collision probability between the pair by +50%.

7.7 Staff, HQ, finances, sponsors, board

Staff roles: Team Principal (you), Technical Director (affects max R&D budget utilization), Chief Designer (aero branch research rate), Chief Engineer (powertrain), Sporting Director (pit crew training), Race Strategist (strategy AI quality in auto-pit), Mechanics pool (pit stop time, reliability fixes). Each has stats and salary.

HQ buildings (each Tier 1–5): HQ Main (admin efficiency), Wind Tunnel (ATR hours), CFD Cluster (ATR hours), Factory (build speed), Design Centre (design capacity), Simulator (driver training), Driver Academy (youth feedback), Hospitality (sponsor appeal), Pit Crew Training (stationary time), Medical Centre (injury recovery), Marketing (sponsor deal flow).

Finance: income = prize money (P1 $50M/year → P10 $5M/year constructor payout) + sponsors (tiered) + engine supply fees (if works team); expenses = salaries + R&D + travel + facility maintenance + build costs. Cost cap enforced separately. **Sponsors** have tier/term/goals/bonus structure: Title sponsor ⩾ $20M with season-finish-position goals; Major $5–20M with race-by-race engagement goals; Minor < $5M fill slots. Goals missed reduce payouts 20%; exceeded trigger renewal-with-bonus.

Board: 4 expectation attributes (results, brand, budget, driver development) each with tolerance bands; quarterly review; firing trigger at 2 consecutive failed reviews with grace period in season 1.

7.8 Championship structure, PU allocation, parc fermé, stewarding, driver market

Multi-tier: **GP3 International** (9 races, generic spec chassis, entry tier) → **Formula Challenge** (12 races, constrained budget, midfield tier) → **Apex Championship** (top tier, 20 races, full systems). Owners can own junior teams and promote drivers. Promotion requires Super Licence points (40 over 3 years) mirroring real regs.

PU allocation per season: 4 ICE, 4 TC, 4 MGU-H, 4 MGU-K, 2 ES, 2 CE. Exceeding triggers grid penalties per Part 3. **Parc fermé** from Q1 start; no setup changes into race. **Race director**: SC/VSC/red flag logic per Part 3; track-limits three-strike system with lap deletion + 5s penalty; penalty ladder 5s/10s/DT/SG/grid/DSQ. **Penalty points** 12 = 1-race ban.

Driver market: contracts 1–3 years; silly season starts May with rumor system, announcements cluster July–September. **Poaching mechanic**: AI teams poach your drivers too (fixes F1M asymmetry). Gardening leave for mid-season switches. Pay drivers bring $5–25M sponsorship. Junior academy pipeline: your Academy building tier gates how many juniors you can develop.

7.9 Pre-season chassis design, livery editor, narrative events, achievements, tutorial

Pre-season is a 6-week mini-campaign: set design philosophy (high-downforce / balanced / low-drag), allocate wind tunnel and CFD hours, run correlation tests, finalize chassis. Design philosophy carries season-long cost-cap and performance implications.

Livery editor: layer-based with primary/secondary/accent colors, 40+ patterns (stripes, splits, fades, geometric), sponsor placement (size + position + rotation for each contract), helmet and suit matching, share-to-cloud with code. Free for base; premium patterns in Battle Pass.

Narrative event system ingests current state (standings, rivalries, morale, calendar) and a seeded event pool to generate 3–7 news/inbox items per race weekend. Event types: rumor, quote, media appearance, rivalry flare, sponsor request, paddock leak, technical protest, board memo. Each has player response options affecting stats/morale/money.

Achievements: 120+ covering onboarding, seasonal milestones, career arcs, hidden easter eggs. **Tutorial**: 3-minute FTUE drops player into a tutorial race weekend after team selection, with contextual tooltips that vanish after step 3, guaranteed first-race win (rigged matchmaking), first upgrade purchase, and first driver hire unlock by end of session 1.

PART 8 — Claude Code prompt library

8.1 CLAUDE.md (repository root)

```markdown
```

Apex GP — Agent Instructions

Mission

Build Apex GP, an unlicensed F1 management mobile game, in React 18 + TypeScript + V

Non-negotiables

- **Deterministic simulation**. Never call `Math.random`. Use the seeded Rng from `s
- **TypeScript strict mode**. No `any` without `// @reason:` comment.
- **No F1/FIA/FOM trademarks** anywhere: teams, drivers, circuits, sponsors, supplie
- **Accessibility**: every interactive element has keyboard/focus handling; reduced-
- **Performance budget**: first contentful paint ≤ 1.5s on mid-tier Android; race ti
- **Ethical monetization**: no pay-to-win, no hidden odds, published drop rates, sof

Stack invariants

- React 18.3 (not 19)
- Motion imported from `motion/react`, never `framer-motion`
- Zustand 5: multi-value selectors require `useShallow`
- Immer requires `enableMapSet()` called once at boot
- Filesystem for save blobs, Preferences for metadata, no localStorage

Conventions

- Files: kebab-case; components: PascalCase
- Exports: named; default only for route-level screens
- Tests: Vitest + @testing-library/react; colocate as `*.test.ts(x)`
- Commits: Conventional (`feat(sim): …`, `fix(ui): …`)

Definition of done

- TypeScript compiles with no errors
- All tests pass
- New code is covered ≥ 80%
- LEARNINGS.md updated if a non-obvious decision was made
- Screenshots updated in `docs/screens/` for UI changes

Directory layout

```
src/
  sim/          — deterministic simulation engine (no React)
    rng/
    tyre/
    weather/
    race/
    season/
  store/        — Zustand slices
  schema/       — Zod save schemas & migrations
  ui/           — React components
    screens/
    primitives/ — shadcn-generated
```

```
    motion/
  services/     — Capacitor integrations (save, purchase, analytics)
  canon/        — fictional teams/drivers/circuits data
  assets/       — static content
```

8.2 LEARNINGS.md starting content

```markdown
# Learnings

Non-obvious decisions and gotchas discovered during implementation. Append-only.

## 2026-04-17 — Stack version pins
- React 18.3 chosen over 19 because Motion 12 ecosystem stability issues on iOS 14 W
- Zustand 5 breaking change: destructuring in selectors without `useShallow` causes
- Vite 6 chosen over 7-alpha (Rolldown) for production stability.
- Capacitor 7.6 (not 8-alpha): SPM isn't required until 8; stick with CocoaPods for

## 2026-04-17 — Simulation determinism
- `sfc32` chosen over `mulberry32`: better statistical quality (passes PractRand), 4
- Seeds are strings hashed via xmur3; splitting streams per subsystem (`race`, `ai`,
- ESLint `no-restricted-properties` banning `Math.random` catches regressions at lin

## 2026-04-17 — Zustand 10k-entity pattern
- Entities as `Map<Id, Entity>` + parallel `Id[]` for iteration order.
- Row components subscribe only to their own id, wrapped in `React.memo`. Bulk tick
- Hot sim loops bypass Immer (mutate Map in place, return `{drivers: state.drivers}`

## 2026-04-17 — Save pipeline
- Hybrid: Filesystem for blob, Preferences for slot metadata.
- Atomic write: copy-to-bak, write-tmp, rename.
- Zod schemas versioned `_version: 7 as const`; migrations array applied sequentiall
```

8.3 The 24-prompt Claude Code sequence

Each prompt is designed to be pasted verbatim into Claude Code running Sonnet 4.5+. Prompts are ordered so each builds on prior output; each has a Definition of Done with tests.

PROMPT 01 — Project scaffold and tooling

> Set up a new Capacitor 7 + Vite + React 18 + TypeScript monorepo at the current directory root. Use Tailwind v4 with Lightning CSS, ESLint (including a `no-restricted-properties` rule banning `Math.random`), Prettier, Vitest with @testing-library/react and jsdom, and strict

> TypeScript. Install dependencies: `react@^18.3`, `react-dom@^18.3`, `typescript@^5.5`, `vite@^6`, `@vitejs/plugin-react`, `tailwindcss@^4`, `zustand@^5`, `immer`, `motion`, `zod@^4`, `@tanstack/react-virtual@^3`, `@capacitor/core@^7`, `@capacitor/cli@^7`, `@capacitor/ios@^7`, `@capacitor/android@^7`, `@capacitor/preferences`, `@capacitor/filesystem`, `@capacitor/status-bar`, `@capacitor/splash-screen`, `@capacitor/haptics`, `@capacitor/app`, `@revenuecat/purchases-capacitor`. Add `capacitor.config.ts` for iOS bundleId `gg.apexgp.app`, Android `gg.apexgp.app`, app name "Apex GP". Create the directory layout from CLAUDE.md. Commit CLAUDE.md and LEARNINGS.md from the spec. Add `npm run dev`, `build`, `test`, `lint`, `cap:sync`, `cap:ios`, `cap:android` scripts. DoD: `npm install && npm run build && npm test` succeeds; empty App component renders "Apex GP" in a Tailwind-styled h1; ESLint fails when `Math.random` is added anywhere.

PROMPT 02 — Seeded RNG and determinism guardrails

> Implement the deterministic RNG layer in `src/sim/rng/`. Create `sfc32.ts` with `sfc32(a,b,c,d)` returning a `() => number` uniform in [0,1), plus `xmur3(seed: string)` hash. Export `createRng(seed: string): Rng` and `makeStreams(masterSeed: string)` returning `{race, ai, damage, events, narrative}`. Implement serialization: `serializeStream(rng)` returns `[a,b,c,d]`, `deserializeStream([a,b,c,d])` returns a primed Rng. Expose utility helpers `uniform(rng, a, b)`, `gaussian(rng, mu, sigma)` (Box-Muller), `pickWeighted<T>(rng, items: Array<{w: number, v: T}>)`, `shuffle<T>(rng, arr)` (Fisher-Yates). DoD: unit tests prove (a) same seed → identical 10k-sample sequence across runs, (b) serialize→deserialize→continue produces identical stream, (c) gaussian samples have μ ≈ mu and σ ≈ sigma within 2% over 10k samples. No `Math.random` anywhere; ESLint passes.

PROMPT 03 — Canon data definitions

> Implement the fictional universe as typed const data in `src/canon/`. Create TypeScript types for `Team`, `Driver`, `Circuit`, `Sponsor`, `Supplier` as specified in the GDD Part 6. Create `teams.ts` with all 10 teams, `drivers.ts` with all 20 drivers (8 stats + meta), `circuits.ts` with all 10 circuits (length, laps, drsZones, scProb, rainPrior, circuitWeights for {chassis,engine,aero,brakes,reliab}), `sponsors.ts` with 40+ fictional sponsors across Title/Major/Minor tiers with goals and payouts, `suppliers.ts` with 4 engine manufacturers, 2 tyre suppliers, 3 fuel suppliers. Export a `CANON` object aggregating all. Add Zod schemas validating the canon. DoD: `CANON` validates under Zod at module-load; unit tests verify counts (10 teams, 20 drivers, 10 circuits, ≥40 sponsors); each driver has a unique id; each team has exactly 2 drivers referenced by id.

PROMPT 04 — Core domain types and constants

> Create `src/sim/types.ts` with TypeScript interfaces for `RaceState`, `DriverRaceState` (position, gap, lap, tyreCompound, tyreAge, fuelKg, ersSoc, mode, morale, dnf), `WeatherState`, `SafetyCarState`, `PitStopOutcome`, `TrackParams`, `CompoundParams`. Create `src/sim/constants.ts` exporting the `F1_CONSTANTS` table from GDD §7.4 with fuel,

> ers, drs, pit, sc, tyre, driver, car, dirty, reliab groups. Export `COMPOUND` dict with {C5,C4,C3,C2,C1,I,W} each having `{baseDeg, cliffPoint, cliffFactor, maxLife, warmup, optTemp}` per GDD §7.2. DoD: types compile strict; constants are `as const` and frozen; unit tests assert specific values (C4 baseDeg 0.08, cliffPoint 15).

PROMPT 05 — Tyre model

> Implement tyre degradation in `src/sim/tyre/`. Export `tyreDelta(compound: Compound, age: number, driver: DriverStats, track: TrackParams): number` per GDD §7.2 formula (warmup + baseDeg×age + cliffFactor × max(0, age−cliffPoint)^2 multiplied by Smoothness modifier). Implement `gripFromTemp(T: number, compound: Compound): number` gaussian. Implement `stepTyreThermal(prev: TyreThermalState, inputs: {slipPower, airflow, ambientC}, dt: number): TyreThermalState` with two-state carcass/tread ODE. Implement graining/blistering accumulation when slip exceeds threshold with temperature mismatch. DoD: unit tests verify (a) C5 at age 5 with default driver on default track returns ~0.45s, (b) cliff kicks in sharply past cliffPoint, (c) Intermediate at 55°C returns grip ≈ 1.0, at 30°C grip < 0.7, (d) warmup penalty on lap 0 is ⩾ 0.3s.

PROMPT 06 — Weather Markov chain and drying

> Implement weather in `src/sim/weather/`. Export `stepWeather(prev: WeatherState, rng: Rng, circuitRainPrior: number, dtMin: number): WeatherState` using the 4-state Markov chain from GDD §7.3 with off-diagonal onset scaled by `circuitRainPrior / 0.25`. Implement `stepWetness(prev: number, inputs: {rainIntensity, airC, trackC, windKph, carsOnTrack}, dtMin: number): number`. Implement `compoundCrossover(wetness: number, currentCompound: Compound): Compound | null` returning a recommended compound change (or null if current is still best). Implement `forecast(current: WeatherState, minutesAhead: number, rng: Rng): { probRain: number, uncertainty: number }` with σ = 0.05 + 0.01 × t. DoD: unit tests verify stationary distribution over 100k steps matches expected within 2% for each circuit; wetness heading to 0 under dry conditions and to 100 under heavy rain; crossover recommends Wet above 70% wetness on slicks.

PROMPT 07 — Lap time core formula

> Implement `lapTime(inputs: LapInputs, rng: Rng): number` in `src/sim/race/lap-time.ts` per GDD §7.1. Compose baseLapTime + tyreDelta + fuelDelta + ersDelta + driverDelta + weatherDelta + trafficDelta + drsDelta + noise + mistakeEvent. Use constants from PROMPT 04. `driverDelta(d, track) = (75 - overall) × 0.05` where overall is weighted stat composite. Noise σ = max(0.05, 0.30 × (1 − consistency/100)). Mistake probability 0.002 × (1 − accuracy/100) adding uniform(rng, 1, 4) seconds when fired. DoD: unit tests verify (a) Verstappen-like 95-rated driver in default car + fresh medium tyres + half fuel returns within 0.5s of baseLapTime, (b) consistency 95 driver has noise σ < 0.05s across 10k samples, (c) full fuel (100kg) adds ~3.3s vs empty.

PROMPT 08 — ERS, fuel, DRS models

> Add `src/sim/race/ers.ts` with `stepErs(prevSoc: number, mode: ErsMode, track: TrackParams): {newSoc, dtLap}` per GDD §7.4. Add `src/sim/race/fuel.ts` with `stepFuel(prevKg: number, mode: FuelMode, track: TrackParams): {newKg, dtLap, wearDelta}` per §7.4 with burn multipliers. Add `src/sim/race/drs.ts` with `drsDelta(tracking, track)` honoring 1.0s detection gap, post-lap-2 rule, SC disable, stacked-diminish 0.85, per-zone 0.15s. DoD: unit tests verify ES saturation at 4 MJ, overtake mode depletes in one lap, DRS returns 0 when gap 1.5s, returns -0.15 at gap 0.5s with 1 zone and -0.28 with 2 zones.

PROMPT 09 — Safety car and pit stop models

> Add `src/sim/race/safety-car.ts` with `stepSafetyCar(state, rng, track)` computing per-lap SC hazard `h = 1 - (1 - P_race)^(1/N_laps)` from per-circuit SC probability and crash-triggered SC (guaranteed on collision). Include VSC and red-flag branches with durations per §7.4. Add `src/sim/race/pit-stop.ts` with `pitStationaryTime(team, rng)`: gaussian(team.pitMuSec, 0.30) with 0.5% wheel-gun failure (+3–10s) and 0.2% cross-thread (+10–30s), floor at 1.8s. `pitLoss(state, track)` = stationary + pitLaneDelta × scFactor (SC 0.40, VSC 0.55, else 1.00). Double-stack adds 3–5s to second car. DoD: Monte Carlo 10k runs: pit stationary mean within 0.05s of mu, σ within 0.05s of 0.30; SC triggers at expected frequency per circuit; double-stack penalty applied only when same team pits same lap.

PROMPT 10 — Overtaking, reliability, race simulation loop

> Add `src/sim/race/overtake.ts` with `overtakeProb(att, def, track, state)` logistic per GDD §7.11 research. `dirtyAir(gapSec, track)` returning 0 above 2s, peaking near 0.3s. Resolve duel outcomes {success, defended, collision, off-track} via rng. Add `src/sim/race/reliability.ts` with per-component per-lap failure sampling using Weibull hazard (k=2.5) scaled by km-used/km-design-life. Assemble `src/sim/race/engine.ts` exporting `simulateRace(config, streams)`: initializes race state, steps lap-by-lap through all drivers, applying lap time, weather, SC/VSC, pit decisions, overtakes, reliability, until all finished or lap count reached. Return `RaceResult` with per-driver positions, lap times, incidents. DoD: deterministic — same seed produces identical result; integration test runs 10-race season, validates (a) no position ties, (b) DNF rate 8–12% across races, (c) overtake count 25–40/race on a 2-DRS circuit.

PROMPT 11 — Season simulation and AI strategy

> Add `src/sim/season/` with `simulateSeason(state, streams)` running the full calendar. Add `src/sim/race/ai-strategy.ts` implementing an AI race strategist: on each lap per driver, decides pit-yes/no, compound choice, fuel mode, ERS mode. Use a scoring heuristic: compute current projected race time, simulate alternative strategy forward 5 laps, pick minimum. Include undercut/overcut awareness (detect rival pit probability), SC window detection, fuel margin check. Add `aiDevelopTeams(season, rng)`: AI teams spend budget on R&D each month, with a target-threshold algorithm that closes gap to leader without overshooting. DoD: seasonal test — AI teams' final Constructor gap to leader is within ±20%

> of expected distribution; AI pit decisions under SC trigger correctly ≥80% of the time; no AI double-stacks in final 3 laps.

PROMPT 12 — Zod save schema v1 and storage pipeline

> Add `src/schema/save.ts` with `SAVE_VERSION = 1` and `SaveV1` Zod schema covering `{_version, seed, rngStreams, season, teams, drivers, inbox, meta}`. Add `src/schema/migrations.ts` with an empty migrations array ready for future versions. Add `src/services/save.ts` exporting `saveGame(slot, data)` implementing the atomic write pattern (copy to bak, write tmp, rename) using @capacitor/filesystem; `loadGame(slot)` runs migrations sequentially and validates with Zod; `listSlots()` reads Preferences for metadata; `deleteSlot(slot)` removes both. DoD: unit tests round-trip a 2MB save through serialize→save→load→deserialize matching deep equal; corrupting the primary file triggers fallback to bak; missing bak throws CorruptSaveError.

PROMPT 13 — Zustand store with slices

> Create `src/store/` with the ImmerStateCreator helper, then slices: `driversSlice`, `teamsSlice`, `partsSlice`, `raceSlice` (live race state), `seasonSlice` (calendar, standings), `financeSlice`, `uiSlice` (focused screen, modals, toasts), `simSlice` (tick control, fast-forward flag). Use Map-based entity stores with parallel id arrays. Wire middleware `devtools(subscribeWithSelector(persist(immer(...))))`. `persist` partializes to exclude `uiSlice` and `simSlice.tickBuffer`. Call `enableMapSet()` at module load. Export `useGameStore` and typed selectors: `selectDriver(id)`, `selectTeam(id)`, `selectStandings()`. DoD: unit tests verify (a) updating one driver re-renders only its subscriber (track via custom subscribe), (b) persist rehydrates state on reload, (c) multi-value selector with `useShallow` does not cause infinite re-render.

PROMPT 14 — App shell, routing, and theme

> Generate shadcn/ui primitives for Button, Card, Dialog, Tabs, Toast, Sheet, Progress, Badge, Avatar with the Tailwind v4 tokens. Set up `src/ui/screens/` with stub screens `Dashboard`, `RaceWeekend`, `Garage`, `Drivers`, `Staff`, `HQ`, `Finance`, `Calendar`, `Inbox`, `Settings`. Implement a bottom tab navigator with five main tabs (Dashboard, Race, Team, HQ, Inbox) and swipeable secondary drawer. Wire `MotionConfig` at the app root, `LazyMotion` with `domAnimation` features. Handle Capacitor StatusBar and SplashScreen on first paint. Support iOS and Android safe-area insets via Tailwind v4 env() utilities. DoD: app builds, all screens render on iOS simulator and Android emulator, tab navigation works with hardware back button on Android, reduced-motion OS setting is respected.

PROMPT 15 — Dashboard and inbox screens

> Build Dashboard screen showing team name/livery, current season standings (constructor P{n} and driver P{n}), next race card (circuit, date, weather forecast with uncertainty), board confidence meter, cash balance vs cost cap utilization, morale of your two drivers, top-3 inbox items. Build Inbox as a virtualized list using @tanstack/react-virtual for 200+ items, grouped by day, with item types {rumor, quote, media, rivalry, sponsor, leak, protest, board}. Each item

> is a Motion-animated Card with gesture-driven archive/star actions. DoD: Dashboard updates on store change without full re-render (profile with React DevTools); Inbox smoothly scrolls 500 items at 60fps on a 2022 mid-tier Android in Chrome DevTools throttling; archiving an item animates out with Motion layout.

PROMPT 16 — Race Weekend screen and live race HUD

> Build the Race Weekend multi-phase UI: Practice (FP1/FP2) → Qualifying (Q1/Q2/Q3) → Warm-up → Race. For each phase, show the track map SVG with dots or cars (live positions), timing tower (P1–P20 with gap), driver radio strip (contextual messages from the narrative engine), strategy panel (current plan, tyre life, fuel margin, ERS SoC bar). Implement playback controls (pause, 1x, 2x, 4x, 8x) that scale sim tick frequency. Use `useMotionValue` for car positions on the SVG to avoid React re-renders. Use Zustand `subscribeWithSelector` to feed motion values directly from the race slice. DoD: a 50-lap race runs at 8x in under 2 minutes without frame drops; pit wall can issue mode changes (ERS, fuel, driving mode) that affect the next lap's result; the timing tower updates on every lap crossing without full re-render.

PROMPT 17 — Garage, livery editor, parts and R&D

> Build Garage screen with sub-tabs: Car Performance (aggregate + per-subsystem bars), Parts (6 part slots with current tier, perk indicator, build-in-progress timer), R&D Tech Tree (three branches with 30 nodes each, unlocked/locked state, cost, risk), Livery Editor (layered canvas editor with primary/secondary/accent color pickers, 40 base patterns, sponsor placement with drag-resize-rotate, helmet/suit matching, export-code-share-cloud). Use Motion `LayoutGroup` for smooth tab transitions; use HTML Canvas or SVG for livery render with real-time preview. DoD: part perks display clearly (regression fix vs MM4); livery editor preserves state across navigation; 5 preset liveries + 40 patterns available; exporting a livery produces a deterministic 12-char share code.

PROMPT 18 — Drivers, staff, HQ screens

> Build Drivers screen: your roster + reserve + market. Each driver has a detailed stats radar, morale meter, rivalry map, contract terms (years remaining, salary, performance bonuses), recent results. Implement contract negotiation dialog with reasonable mid-fidelity (counter-offer buttons, personality-driven reactions). Build Staff screen with the 7 roles, hire/fire, stat ranges, salaries. Build HQ screen with 11 buildings, each with Tier 1–5 upgrade path, cost/time/effect preview, "start construction" action, queue visualization. Use Motion `layoutId` for shared-element transitions between list and detail. DoD: driver market refreshes weekly with new candidates including pay drivers, veterans, and rookies; staff hires respect cost cap exclusions; HQ construction progresses on season tick.

PROMPT 19 — Finance, calendar, and cost cap enforcement

> Build Finance screen with income breakdown (prize money, sponsors, engine supply), expense breakdown (salaries, R&D, build costs, travel, facilities), cost cap utilization gauge with FIA audit warning at 95%+, sponsor management (current deals with goals/progress,

> incoming offers, renewal reminders). Build Calendar screen with the 10-race season, results per round, expandable per-round Race Replay. Implement cost cap enforcement: attempts to build parts that would exceed cap are blocked; mid-season FIA audit fires with 10% probability per month, with detection probability proportional to overspend magnitude, triggering grid penalties or fines. DoD: finance totals match ledger of all recorded transactions to the dollar; cost cap audit fires deterministically from RNG stream; sponsor renewals offer bonuses for exceeded goals, penalties for missed.

PROMPT 20 — Narrative event engine

> Implement `src/sim/narrative/` producing 3–7 inbox items per race weekend. Build event templates per type (rumor, quote, media, rivalry, sponsor, leak, protest, board). Each template has trigger conditions (state predicates), content generators (seeded text variants), and player response options affecting stats/morale/money. Implement rivalry state machine: rivalry forms when 2 drivers collide or overtake ⩾3 times in 5 races; stored as symmetric `rivalryMatrix<DriverId, DriverId, intensity>`. Implement team orders events with driver response probability scaled by Independence trait and current morale. DoD: a simulated 10-race season generates 40+ distinct inbox items; rivalries form/fade tracked across seasons; player response changes measurable downstream state.

PROMPT 21 — Battle Pass, store, and RevenueCat

> Wire RevenueCat configure with iOS and Android public API keys, appUserID stable across reinstalls. Define offerings: `racing_pass` (entitlement `pass_standard`), `championship_pass` (entitlement `pass_championship`), `tokens_tier1` through `tier5` (consumables), `starter_pack_72h` (non-consumable first-time unlock), `remove_ads` (non-consumable), `livery_packs` (non-consumable). Build Store screen with packages pulled from `getOfferings()`, prices localized automatically. Build Battle Pass screen: free + premium tracks, 50 tiers each, XP progression via race weekends, cosmetic rewards (liveries, helmet patterns, garage skins). Published drop rates shown adjacent to any random-reward offers. Soft pity at 70% progress, hard pity at 80 pulls. Server-validate tokens on backend by RevenueCat webhook. DoD: TestFlight sandbox purchase completes and unlocks entitlement; Restore Purchases button works; consumable tokens sync via backend; drop rates displayed on every random reward surface.

PROMPT 22 — Analytics, live-ops calendar, push notifications

> Integrate GameAnalytics (or Firebase Analytics) via Capacitor plugin. Emit typed events per spec: session_start, race_completed, upgrade_purchased, iap_started, iap_completed, funnel_onboarding, save_corrupted. Implement a client-side live-ops calendar keyed to weekly and monthly cadences with 72-hour Race Weekend Events that match abstract round numbers to real F1 weekends via a calendar config file updateable without app update (fetched from a CDN on app open, with local fallback). Implement push notifications via @capacitor/push-notifications (Firebase/APNS): delayed pre-permission screen triggered only after first-race-win; frequency cap 1/day + 5/week; context-aware messages ("Race starts in 1h", "Pit crew finished", "Battle Pass ends in 48h"); segmented by D3/D7/D14/D30 lapse. DoD: events land in GameAnalytics dashboard; opt-in prompt appears only after win;

> notification scheduler respects frequency caps; live-ops config fetches on cold start without blocking.

PROMPT 23 — Performance hardening and 60fps race view

> Profile the race view with React DevTools and Chrome DevTools. Target: 20-driver race tick executes in ≤16ms on a 2022 mid-tier Android. Implement: (a) replace Immer in the hot race tick path with direct Map mutation, (b) move car-position rendering to `useMotionValue` bypassing React, (c) virtualize any list >50 items, (d) strip all `layout` props from race-tick components, (e) switch to `LazyMotion` with `domAnimation` features only, (f) manual chunk split react/motion/ui, (g) audit `useMemo`/`useCallback` for net benefit and remove those with negative ROI (or enable React Compiler opt-in). DoD: Chrome DevTools Performance trace of a 3-minute race at 4x shows no frame drops below 58fps; bundle analyzer shows critical chunk ≤500KB gzipped; Safari Web Inspector on iOS device shows equivalent perf.

PROMPT 24 — Submission hardening and launch readiness

> Prepare for store submission. iOS: add `PrivacyInfo.xcprivacy` declaring NSPrivacyAccessedAPITypes with reason codes CA92.1 (UserDefaults), C617.1/0A2A.1 (file timestamp); add NSPhotoLibraryUsageDescription and other Info.plist keys only for actually-used capabilities; **do not** include ATT framework unless ad SDKs require it. Android: upgrade targetSdk to 35, compileSdk to 35, set Play Billing 7; complete Data Safety form matching SDK declarations; enable Play Integrity; prepare closed-testing track with 12 testers opted-in for 14 days if developer account is new. Both: host privacy policy and ToS; implement in-app "Delete Account" path; Restore Purchases button in Settings; age rating 12+ (virtual currency + fictional violence/collisions minimal). Implement OOBE tutorial per GDD §7.9: 3-minute FTUE with guaranteed first-race win, first upgrade, first driver hire by end of session. DoD: TestFlight build uploads without rejection warnings; Play Console internal testing track accepts build; onboarding analytics funnel shows 80%+ step-3 retention in internal tests; privacy manifest validator tool reports zero issues.

Conclusion

Apex GP's strategic opening is wider than any F1 management release since 2016. **Frontier's F1 Manager exit leaves the licensed market empty**; Motorsport Manager 4 is widely panned as a "carbon copy" abandoned after a year; F1 Clash is under ASA enforcement pressure for loot-box practices and players are churning at the Series 5–6 pay wall; iGP Manager's UI limits its reach; Motorsport Manager's spiritual sequel is unconfirmed. **An unlicensed, deeply simulated, ethically monetized, narrative-rich F1 management game with proper multiplayer hooks and mod-like content flexibility has no direct competitor.** The core design insight from all this research is that the hardcore-versus-casual split that killed Frontier's F1M is resolvable on mobile by pacing — the same 5-minute race-weekend session as F1 Clash, but layered over a Motorsport Manager-PC-grade simulation underneath — with a Drive-to-Survive-style narrative engine as the flywheel that connects them.

Three non-obvious bets define this production. **First**, determinism is the foundation — seeded RNG streams, Zod-validated saves, migration-safe schemas — because it enables Ghost Races tied to real F1 weekends and because it is a prerequisite for any future multiplayer or replay system without rewriting the engine. **Second**, the monetization rails are cosmetic-biased from day one — livery editors, helmet packs, garage skins, Battle Pass — because an unlicensed game cannot lean on real-driver card pulls for revenue and because ASA enforcement is now a real regulatory risk. **Third**, the GDD treats features fans have begged for over a decade (livery editor, F2/F3 academy pipeline, historical seasons, proper cost cap, power unit allocation, press conferences, mod-like content flexibility) as **launch-scope requirements, not post-launch ambitions**, because every one of those features exists as a testable formula or a tractable data structure in Part 7. The 24-prompt Claude Code sequence in Part 8 is ordered so a solo developer working sequentially with Sonnet 4.5+ can ship a soft-launchable build in 14–18 weeks and hit global launch gates in 26–30 weeks. That is the production bible.